



Course Name

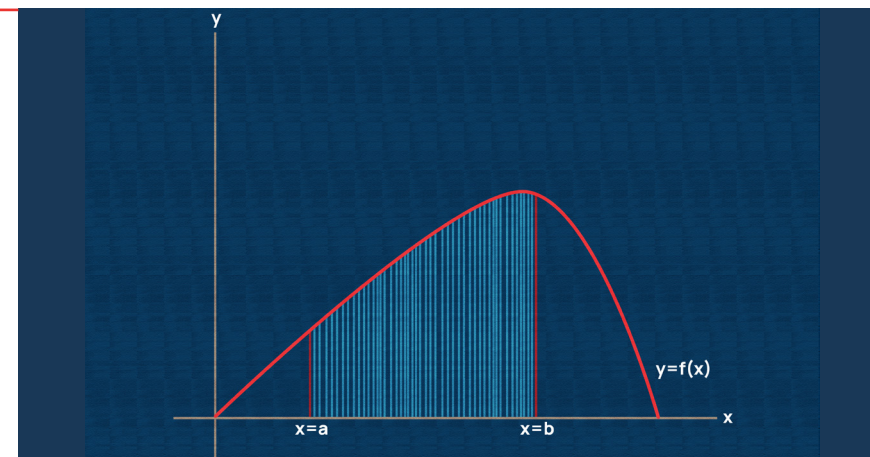
ELEMENTARY MATHEMATICS II (CALCULUS)

Topic: The Area of a Definite Curve

WEEK 10

IN SUMMARY

This week, you learned about definite integrals and their applications in area calculation. Definite integrals enable us to determine the exact area bounded by functions and the x-axis. The key concept is that the integral represents the signed area enclosed between the curve and the x-axis from a starting point 'a' to an endpoint 'b'. We achieved this by breaking down the region into infinitesimally small vertical strips, meticulously calculating each strip's area, and then summing them up. This process provides the net area between the curve and the x-axis. You also learned the concept of Riemann sums, which allow us to calculate the area beneath complex curves. Additionally, you learned how definite integrals can be applied to find the area between two curves and dealt with examples where the curves intersect or have intricate features. Definite integrals prove to be a powerful tool for precisely determining areas between curves, regardless of their complexity.



BULLET POINT SUMMARY

- Definite integrals can be used to calculate the area under a curve, the area between two curves, and the area within polar coordinates.
- Definite integrals can also be used to determine the area of three-dimensional curves and the area of irregular shapes.
- Definite integrals are a fundamental concept in calculus with countless practical implications in fields spanning from physics to engineering and economics.

CASE STUDY

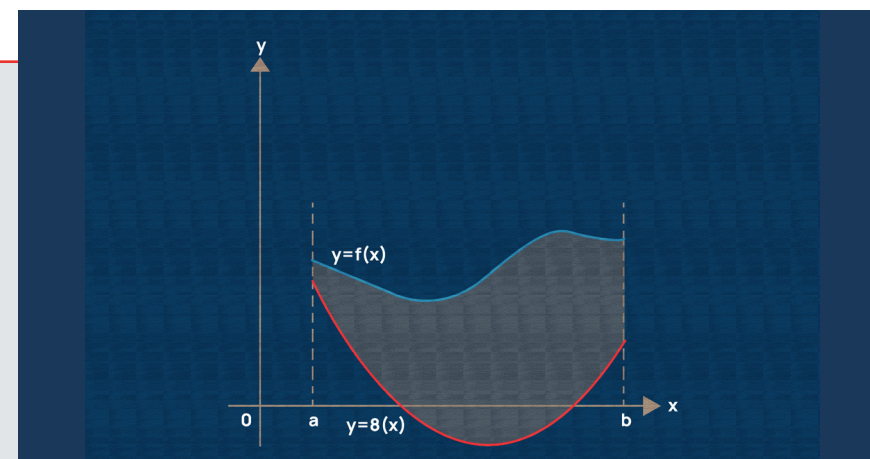
Title: Using Definite Integrals to Calculate the Cross-Sectional Area of a Dam

Introduction

Definite integrals are used in a wide variety of industries to calculate areas such as engineering, physics, and economics. One common application is in calculating the cross-sectional area of a dam. This information is used to determine the dam's strength and stability.

Objective

To demonstrate the application of definite integrals to calculate the cross-sectional area of a dam.



CASE STUDY CONT'D

Methodology

The following steps will be used to solve the case study problem:

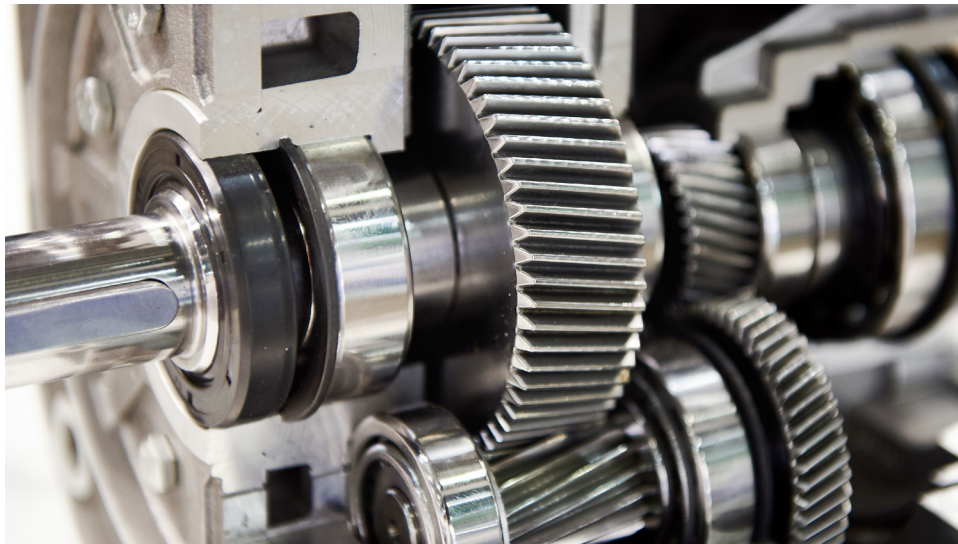
- Identify the function that represents the cross-sectional profile of the dam.
- Determine the interval over which the area is to be calculated.
- Evaluate the definite integral of the function over the interval.
- Interpret the result of the integral as the cross-sectional area of the dam.

Results

The case study will show how definite integrals can be used to calculate the cross-sectional area of a dam. The specific results will depend on the specific dam that is addressed.

QUESTIONS TO PONDER

- What are the different methods for evaluating definite integrals?
- What are the different applications of definite integrals in engineering?
- How can definite integrals be used to solve real-world problems?

**SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON**

After this lesson, you have acquired skills in mathematical analysis, problem-solving, and critical thinking. I have also gained proficiency in applying definite integrals to calculate land areas, making me well-equipped for real-world applications in civil engineering.

PERSONAL REFLECTION

I found this lesson on definite integrals to be very informative and engaging. I am glad that I had the opportunity to learn about this important topic in calculus. I am confident that the skills and competencies I have acquired in this lesson will be valuable to me in my future studies and career.